Measuring the Radius of a Star

Question A.1

Answer	Marks
Along the horizontal stretches the paths differ by $L_1 - L_2$; in addition ray 1 travels the distance L_3 twice. Hence $$\Delta_{12} = L_1 + 2L_3 - L_2.$$	
Answer $\Delta_{12} = L_1 + 2L_3 - L_2$. (If an extra term $\lambda/2$ is written here, or a phase difference is computed instead of the path difference, A2 is marked consistently with this error.)	0.2 pts
Total	0.2 pts

Question A.2

Answer	Marks
The paths of rays 1 and 1′ from M3 to O form an isosceles triangle, and the changes of the horizontal and vertical stretches caused by the shifted point of incidence cancel. The only parameter that changes the path of the ray is the z coordinate of its point on M3, which shortens the stretch from HM to M3; ray 2 is unchanged. Hence $$\Delta_{1'2} = \Delta_{12} - 2(L_3 + L_4)\sin^2\alpha = L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\sin^2\alpha \approx L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\alpha^2.$$	
Shown that the optical paths of rays 1 and 1′ from mirror M3 to the screen are equal.	0.5 pts
Shown that the optical path difference of rays 1 and 1′ from the beam splitter to M3 is $\Delta_{11'} = 2(L_3 + L_4)\sin^2\alpha$ (the small-angle limit was not stated, so only the exact result earns this).	0.5 pts
Answer $\Delta_{1'2} = L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\sin^2\alpha \approx L_1 + 2L_3 - L_2 - 2(L_3 + L_4)\alpha^2$ (exact result required).	0.4 pts
Total	1.4 pts

Question B.1

Answer	Marks
The rays deviate from the horizontal and vertical directions only by small angles, so to first order these deviations do not change the path lengths. Because the beam splitter is at $\pi/4$, a change of the horizontal stretch from M1 (M2) to the beam splitter is exactly compensated by the opposite change of the stretch from the beam splitter to the screen, for both rays.	
Stated that, because the angles to the vertical and horizontal are small, they need not be taken into account in the first-order optical path.	0.2 pts

Used the $\pi/4$ inclination of the beam splitter to show that the changes of the horizontal stretch from M1 (M2) to the beam splitter and of the stretch from the beam splitter to the screen cancel — for ray 1 and for ray 2 (0.3 each).	0.6 pts
Total	0.8 pts

Question B.2

Answer	Marks
Ray 1: its path from M3 to the screen is constant; when the point of incidence on M1 moves, the path from M1 to the beam splitter grows exactly as much as the sum of the paths from the dashed line to M1 and from the beam splitter to M3 shrinks. Ray 2: its path from M2 to the beam splitter is constant, and the sum of the paths from the dashed line to M2 and from the beam splitter to the screen is constant.	
Ray 1: the optical path between M3 and the screen is constant.	0.1 pts
Ray 1: as the point of incidence moves, the path from M1 to the beam splitter grows exactly as the sum of the paths from the dashed line to M1 and from the beam splitter to M3 shrinks.	0.4 pts
Ray 2: the optical path from M2 to the beam splitter is constant.	0.1 pts
Ray 2: the sum of the paths from the dashed line to M2 and from the beam splitter to the screen is constant.	0.2 pts
Total	0.8 pts

Question B.3

Answer	Marks
δ_O is the phase difference at $\theta = 0$ (Part A). Besides the path difference, ray 1 has one more reflection and so an extra π: $$\delta_O = \frac{2\pi}{\lambda}(L_1 + 2L_3 - L_2) + \pi.$$ By B1 and B2, a change of θ changes the phase only through the paths up to the dashed line $z_\mathrm{d} = L_3 + L_4$, which differ by $(L_1 + L_2)\theta$: $\frac{2\pi}{\lambda}(L_1 + L_2)\theta = \frac{2\pi}{\lambda} b\theta$, so $$b = L_1 + L_2.$$	
Answer for $\delta_O = \frac{2\pi}{\lambda}(L_1 + 2L_3 - L_2) + \pi$.	0.2 pts
Stated that the additional phase difference comes from the different optical paths of the rays up to the dashed line.	0.4 pts
Answer $b = L_1 + L_2$.	0.4 pts
Total	1.0 pts

Question B.4

Answer	Marks
Two rays 1 that reach points of the screen a distance y apart were reflected from M3 at points also (for small α) y apart. Their paths differ only through the z coordinate of the point of reflection on M3, which shortens both the stretch from HM to M3 and that from M3 to the screen: $\Delta L = -2\alpha y$, so $\delta_y = \delta_O - \frac{2\pi}{\lambda} \cdot 2\alpha y$.	
Stated that if the rays reach the screen at coordinate y, the point of incidence of ray 1 on M3 also shifts by y.	0.3 pts
Shown that the optical path of ray 1 from the beam splitter to the screen decreases by $2\alpha y$.	0.3 pts
Total	0.6 pts

Question B.5

Answer	Marks
With the numbers, $\delta = 81\pi - 800\pi\, y/(1\text{ m})$. Maxima are at $\delta = 2\pi n$; point O is a minimum. Hence $$y_1 = \frac{\lambda}{4\alpha} = 1.25\text{ mm}, \qquad \Delta y_m = \frac{\lambda}{2\alpha} = 2.5\text{ mm}.$$	
Answer $\Delta y_m = 2.5$ mm (expression and value, 0.1 each).	0.2 pts
Answer $y_1 = 1.25$ mm (marked even with the wrong sign).	0.4 pts
Total	0.6 pts

Question B.6

Answer	Marks
Ray 1 meets the beam splitter twice (reflected, then transmitted), ray 2 once, so the field amplitude of ray 1 is $\sqrt{2}$ times smaller and $I_1 = I_2/2$. Then $I \propto 1 + 2 + 2\sqrt{2}\cos\delta$ and $$I(y) = I_0\,\frac{3 - 2\sqrt{2}\cos\left[80\pi\,(1 - 10\,y/(1\text{ m}))\right]}{3 + 2\sqrt{2}}.$$	
Formula $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$.	0.2 pts
Taken into account that $I_1 = I_2/2$.	0.1 pts
Formula $I(y) = I_0\,(3 - 2\sqrt{2}\cos\delta(y))/(3 + 2\sqrt{2})$.	0.3 pts
$\delta(y) = \pm 80\pi\,(1 - 10\,y/(1\text{ m}))$.	0.3 pts
Total	0.9 pts

Question C.1

Answer	Marks
For small θ_x, θ_y: $k_z = -k$, $k_x = -k\theta_x$, $k_y = -k\theta_y$, i.e. $$\vec{k} = -k\left(\theta_x\hat{x} + \theta_y\hat{y} + \hat{z}\right).$$	
Taken into account that every projection is negative, $\lvert k_i\rvert = -k_i$.	0.1 pts
Magnitudes of the projections $\lvert k_x\rvert = k\theta_x$, $\lvert k_y\rvert = k\theta_y$, $\lvert k_z\rvert = k$ (0.1 each).	0.3 pts
Total	0.4 pts

Question C.2

Answer	Marks
The initial offset along x of rays that meet at one point of the screen is of order $L_3\theta_x$, so the path difference it causes is of order θ_x^2 and θ_x drops out. As in Part B: $$\delta_{1''2''} = \frac{2\pi}{\lambda}\left((L_1+L_2)\theta_y + (L_1+2L_3-L_2) - 2\alpha y\right) + \pi.$$	
Shown that the phase difference does not depend on θ_x.	0.3 pts
Answer $\delta_{1''2''} = \frac{2\pi}{\lambda}\left(L_1+2L_3-L_2+(L_1+L_2)\theta_y - 2\alpha y\right) + \pi$.	0.3 pts
Total	0.6 pts

Question D.1

Answer	Marks
The parts of the star radiate independently, so their intensities add. With mirrors moved apart by d (Part D below; $d = 0$ here), a patch contributes $$\mathrm{d}I = \mathcal{J}\,\mathrm{d}\theta_x\mathrm{d}\theta_y\left(3 - 2\sqrt{2}\cos\left(\varphi_0 - \tfrac{4\pi}{\lambda}\alpha y + \tfrac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)\right), \quad \varphi_0 = \tfrac{2\pi}{\lambda}(L_1+2L_3-L_2).$$ Integrating over θ_x with the hint, $I_{\max,\min} = \int_{-\theta_R}^{\theta_R}\mathcal{J}\sqrt{\theta_R^2-\theta_y^2}\left(3 \pm 2\sqrt{2}\cos\tfrac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)\mathrm{d}\theta_y$ (twice, up to a common factor), so $$V = \frac{2\sqrt{2}\int_{-\theta_R}^{\theta_R}\sqrt{\theta_R^2-\theta_y^2}\cos\left(\tfrac{2\pi}{\lambda}(L_1+L_2+d)\theta_y\right)\mathrm{d}\theta_y}{3\int_{-\theta_R}^{\theta_R}\sqrt{\theta_R^2-\theta_y^2}\,\mathrm{d}\theta_y}.$$ With $w = \theta_y/\theta_R$ the denominator integral is $\pi/2$; with $\eta = \tfrac{2\pi}{\lambda}(L_1+L_2+d)\theta_R$ and the even integrand, $$V = \frac{8\sqrt{2}}{3\pi}\int_0^1\sqrt{1-w^2}\cos(\eta w)\,\mathrm{d}w.$$	

Correct contribution $\mathrm{d}I$ of a small patch of the star.	0.3 pts
Intensity at an arbitrary point as an integral over one variable.	0.3 pts
Correct $I_{\max}$ and $I_{\min}$ (0.3 each).	0.6 pts
Formula for the visibility V as a ratio of integrals.	0.4 pts
Denominator evaluated and final answer $V = \frac{8\sqrt{2}}{3\pi}\int_0^1 \sqrt{1-w^2}\cos(\eta w)\,\mathrm{d}w$.	0.2 pts
Total	1.8 pts

Question D.2

Answer	Marks
The first zeros of the visibility are η_1, η_2 for $n = 0.5$. Hence $\eta_2 - \eta_1 = \frac{2\pi}{\lambda}(d_2 - d_1)\theta_R$, $$\theta_R = \frac{(\eta_2 - \eta_1)\lambda}{2\pi(d_2 - d_1)} = 1.18 \cdot 10^{-7}, \qquad R = \theta_R h = 7.16 \cdot 10^8 \text{ km}.$$	
Stated that the first zeros of the integral are η_1, η_2 for $n = 0.5$.	0.4 pts
Correct formula $\theta_R = (\eta_2 - \eta_1)\lambda/(2\pi(d_2 - d_1))$.	0.3 pts
Correct value $\theta_R = 1.18 \cdot 10^{-7}$.	0.1 pts
Correct radius $R = 7.16 \cdot 10^8$ km.	0.1 pts
Total	0.9 pts